De Moivre's theorem

Basic skills

Q1. a) $(\cos x + i\sin 3x)^3 = \cos 3x + i\sin 3x$

b) $\cos 6x + i\sin 6x = (\cos x + i\sin x)^6$

c) $(\cos x + i\sin x)^{-5} = \cos(-5x) + i\sin(-5x)$

$= \cos(5x) - i\sin(5x)$

Q2. By c) $\cos(5x) - i\sin(5x)$

$= \cos(-5x) + i\sin(-5x)$

$= (\cos x + i\sin x)^{-5}$

Q3. a) $\left(\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right)^3$

$= \cos(2\pi) + i\sin(2\pi)$

$= 1 + i\times 0 = 1$

b) $\left(2\cos\left(\frac{\pi}{4}\right) + 2i\sin\left(\frac{\pi}{4}\right)\right)^4$

$$= 2^4\left(\cos\left(\tfrac{\pi}{4}\right) + i\sin\left(\tfrac{\pi}{4}\right)\right)^4$$

$$= 16\left(\cos\pi + i\sin\pi\right)$$

$$= 16 \times (-1 + 0i) = -16$$

c) $(2\cos 50° + 2i\sin 50°)^4$

$$= 2^4\left(\cos 200° + i\sin 200°\right)$$

$$= 16\left(-0.9397 + i \times -0.3420\right)$$

$$= -15.04 - 5.47i$$

Competency skills

Q4. a) If $z = \cos x + i\sin x$

then $\dfrac{z + z^{-1}}{2} = \cos x$

z^{-1} is z^* in this case

so $\cos^3 x = \left(\dfrac{z + z^{-1}}{2}\right)^3$

$$= \frac{1}{8}\left(z + z^{-1}\right)^3$$

b) $\cos^3 x = \frac{1}{8}(z + z^{-1})^3$

$= \frac{1}{8}\left(z^3 + 3z^2 z^{-1} + 3z z^{-2} + z^{-3}\right)$

$= \frac{1}{8}\left(z^3 + 3z + 3z^{-1} + z^{-3}\right)$

c) $\cos^3 x = \frac{1}{8}\left(z^3 + z^{-3} + 3(z + z^{-1})\right)$

$= \frac{1}{8}\left(2\cos 3x + 3 \times 2\cos x\right)$

as $\cos nx = \frac{z^n + z^{-n}}{2}$

$= \frac{1}{4}\cos 3x + \frac{3}{4}\cos x$

Q5. $\sin^5 x = \left(\frac{z - z^{-1}}{2i}\right)^5$

$= \frac{1}{32i^5}(z + z^{-1})^5$

$= \frac{1}{32i}\left(z^5 - 5z^3 + 10z - 10z^{-1} + 5z^{-3} - z^{-5}\right)$

$= \frac{1}{32i}\left(z^5 - z^{-5} - 5(z^3 - z^{-3}) + 10(z - z^{-1})\right)$

$$= \frac{1}{32i}\left(2i\sin 5x - 5 \times 2i\sin 3x + 10 \times 2i\sin x\right)$$

$$= \frac{1}{16}\sin 5x - \frac{5}{16}\sin 3x + \frac{10}{16}\sin x$$

$$= \frac{1}{16}\sin 5x - \frac{5}{16}\sin 3x + \frac{5}{8}\sin x$$

Q6. $((\cos x + i\sin x)^n = (e^{ix})^n$

$= e^{inx} = e^{i(nx)}$

$= \cos(nx) + i\sin(nx)$

Q7. a) $(i-2)^5$

for $z = i-2$ convert form

$|z| = \sqrt{1^2+2^2} = \sqrt{5}$

as $x<0$, $y>0$

$\operatorname{Arg} z = \tan^{-1}\left(\frac{1}{-2}\right) = -0.463 + \pi$

so $\left(\sqrt{5}\left(\cos(-0.463+\pi) + i\sin(-0.463+\pi)\right)\right)^5$

$= (\sqrt{5})^5 \left(\cos(5(\pi - 0.463)) + i\sin(5(\pi - 0.463)) \right)$

calc ↓

$= 37.867 + 41.123i$

b) $(\sqrt{3} + i)^{12}$

$z = \sqrt{3} + i$ $\quad |z| = \sqrt{4} = 2$

$\text{Arg} = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

so $\left(2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right)^{12}$

$= 2^{12}(\cos 2\pi + i\sin 2\pi)$

$= 2^{12}(1 + 0i) = 2^{12} = 4096$

Advanced skills

Q8. a) $\cos^6 x = \left(\frac{z + z^{-1}}{2}\right)^6$

$= \frac{1}{2^6}\left(z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}\right)$

$$= \frac{1}{64}\left(z^6 + z^{-6} + 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) + 20\right)$$

$$= \frac{1}{64}\left(2\cos 6x + 6 \times 2\cos 4x + 15 \times 2\cos 2x + 20\right)$$

$$= \frac{1}{32}\cos 6x + \frac{3}{16}\cos 4x + \frac{15}{32}\cos 2x + \frac{5}{16}$$

b) $\int \cos^6 3x \, dx$

$$= \int \left(\frac{1}{32}\cos 18x + \frac{3}{16}\cos 12x + \frac{15}{32}\cos 6x + \frac{5}{16}\right) dx$$

$$= \frac{1}{32 \times 18}\sin 18x + \frac{3}{16 \times 12}\sin 12x + \frac{15}{32 \times 6}\sin 6x + \frac{5}{16}x$$

$$= \frac{1}{576}\sin 18x + \frac{1}{64}\sin 12x + \frac{5}{64}\sin 6x + \frac{5}{16}x$$

Q9. Because it can be used to write powers of trig as linear multiples instead (or vice-versa)

Q10. a) Proof of $\cos nx = \frac{z^n + z^{-n}}{2}$

$z^n = (\cos x + i\sin x)^n$ by De moivres

$= \cos(nx) + i\sin(nx)$

$z^{-n} = \cos(-nx) + i\sin(-nx)$

$= \cos(nx) - i\sin(nx)$

so $z^n + z^{-n} = 2\cos(nx)$

hence $\cos(nx) = \frac{z^n + z^{-n}}{2}$

b) Proof of $\sin(nx) = \frac{z^n - z^{-n}}{2i}$

By above $z^n - z^{-n} = 2i\sin(nx)$

so $\sin(nx) = \frac{z^n - z^{-n}}{2i}$